Update Conference
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Extensibility in .NET Applications

with WebAssembly

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“The plugin model to end all plugin models”

YAGNI

Are we there yet?

About Me

- My career in movie quotes
  - “The horror! The horror!” (Apocalypse Now)
  - “Nuke it from orbit” (Aliens)
  - “What did you see, old man?” (Godzilla)

- Street creds
  - https://www.nuget.org/packages/Opa.Wasm/ (11/2019)

https://github.com/christophwille/demosatconferences/tree/master/UCP25

SharePoint

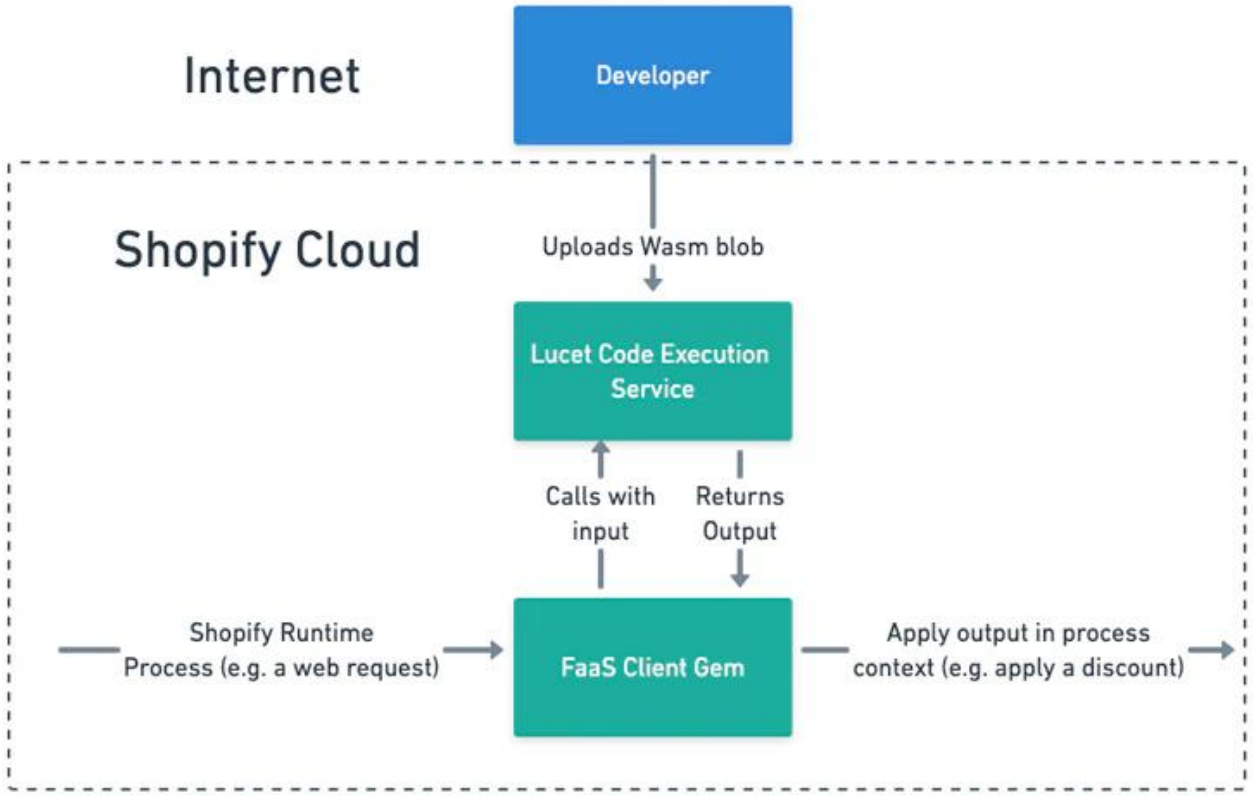
(on premises)

Plugin Considerations

- (Hostile) Multi-Tenancy
  - Performance
  - Isolation
- Multi-language
- Host events, host functions

- plugin, def.: unknown runtime dependency that hooks into the data lifecyle

Shopify



How Shopify Uses WebAssembly Outside of the Browser

https://shopify.engineering/shopify-webassembly#

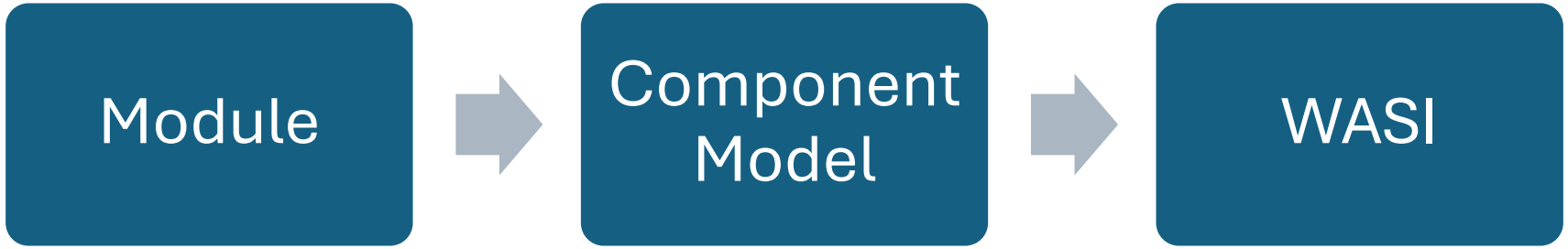
WebAssembly in Very Short

- Checks all the boxes: secure sandboxing, language agnostic, cross-platform, resource control
- Portable compilation target
- Binary instruction format for a stack-based VM
- Control-Flow Integrity (CFI)
- (Very) Limited type system, basically int and float
- Linear memory and streams are your friends (at the low level)

Abstractions

- Do it yourself
- Single language / platform (wasm-bindgen)
- Lightweight framework (Extism)
- Full control (component model)

Abstractions



Extism

- Universal abstraction over WebAssembly
- Plugin consistency across languages
  - CLI, desktop app, TS in VSCode, JetBrains…
- (Wasm) Runtime choice depends on SDK language
- PDK only adds kbyte-sized overhead
- Memory copy needs to be done anyways, simply use JSON
  - "bytes-in → bytes-out" programming model (RPC vs JSON)
- vs Component Model
  - "Getting stuff done“, "Value first, not standards first“
  - Yes, there is overlap (Steve Manuel likens it to jQuery)

Demo time

- Diving into Extism

For Inspiration

Adding Extensibility to Twenty, a Modern CRM

https://www.getxtp.com/blog/adding-extensibility-to-twenty-a-modern-crm

Build moss extensions more easily

https://github.com/RedTTGMoss/Moss.NET.SDK/

Component Model & .NET

- Mono & native AOT are community supported, early stages
- You don’t want to manually do all the steps
- You want
  - https://github.com/bytecodealliance/componentize-dotnet

Exploring C# Wasm Components - James Sturtevant, Microsoft (Nov 2024)

https://www.youtube.com/watch?v=cwWb-bE3c4o

Demo time

- WASI & Component Model .NET

Aside: Right Tool for the Job

- Remember the TS compiler in Go controversy?
  - https://devblogs.microsoft.com/typescript/typescript-native-port/
- Remember the multi-language/compilation target support
  - Host != Plugin

Are We There Yet? Revisited

- Wac-ing component + host together doesn't cut it
- You need a dynamic host with Component Model support

https://github.com/bytecodealliance/wasmtime-dotnet/issues/324

- Native runtime would be nice (*)
  - (*) https://github.com/kelnishi/WACS is an interpreter

.NET 10: No WASI

https://henrikrxn.github.io/blog/Wasi-dotnet-10/

WASI Evolution (Are We There Yet? Revisited)

Jan 2024

WASI 0.2.0
- Stable
- Integrated into language SDKs
- Years of support likely

2024

WASI 0.2.1 => 0.2.n
- Additive non-breaking changes
- Regular time-based releases

Mid 2025

WASI 0.3.0
- Support for async
- More ergonomic interfaces

2026???

WASI 1.0
- Stable branch
- No additional breaking changes

Supported worlds on WASI foundations will grow 🌱

- `wasi-cloud`
- `wasi-nn`
- `wasi-tls`
- `wasi-webgpu`

Source:

Wasm I Right or Wasm I Wrong? a Review of the Wasm Ecosystem - Taylor Thomas & David Justice (April 2025)

https://www.youtube.com/watch?v=KK0FKiQ7nis

YAGNI, Revisited

- How about https://github.com/microsoft/RulesEngine
- or https://github.com/sebastienros/jint
  - allows accessing Clr
  - has execution constraints

ADR-Style Summary

- Do you even have hostile multi-tenancy requirements?
  - or an enterprise app that needs extensibility
  - or an enterprise app that needs business rule extensibility
- Does your app need extensibility in multiple languages?
- Is the ecosystem stable?
  - Component model, WASI, Extism, componentize-dotnet, Wasmtime
- Is it sound to depend on?
  - Github stars, actitivity, # of developers?

It depends

(especially when using cloud-native technologies in the .NET ecosystem)